

Marwan Bukhori

Full-Stack Software Engineer

Cyberjaya, Malaysia · +60 13-596 1559 · marwanbukhori.dev@gmail.com
linkedin.com/in/marwanbukhori · github.com/marwanbukhori · marwanbukhori.com

PROFESSIONAL SUMMARY

Full-stack software engineer with 4+ years building scalable backend services, event-driven architectures, and enterprise web applications across fuel retail, e-invoicing, intelligence dashboard and government analytics. Specialized in Typescript, Javascript, Python, Laravel PHP and AWS serverless infrastructure, with production systems serving 1,000+ fuel stations and processing 800K+ daily transactions.

Strong record in CQRS, domain-driven design, and security hardening. Applied harness-engineering and agentic workflow using Claude Code, Git worktrees, Claude's skill & plugin (Superpower) and currently learning Agent development life cycle using Langchain libraries. Open to immediate opening and adjacent AI-driven engineering, platform, DevOps, and data engineering roles.

TECHNICAL SKILLS

Languages	JavaScript, TypeScript, PHP, Python, SQL
Frameworks	NestJS, Next.js, FastAPI, Nuxt.js, Laravel, Django, Vue.js, React Native, Express, Jest, Cypress, TanStack Query, Zustand, AlpineJS, PrimeVue, Tailwind CSS
Cloud & Infrastructure	AWS (Lambda, ECS, SQS, Kinesis, RDS, DynamoDB, ElastiCache, S3, EC2, CloudWatch, WAF), AWS CDK, Serverless Framework, Docker, Kubernetes, GitLab CI/CD, Supabase
Databases	MySQL, PostgreSQL, MongoDB, DynamoDB, Redis
Messaging	AWS SNS, AWS SQS, AWS Kinesis, MQTT, Pub/Sub patterns
Architecture	Microservices, Serverless, Event-Driven, CQRS, Domain-Driven Design (DDD), Multi-tenant, Message Queue patterns
Methodologies	Test-Driven Development (TDD), Agile/Scrum
Tools	Claude, MCP, Langchain (Learning) Git, Worktrees, Jira, Confluence, Postman, Power BI, Google Sheets API

EXPERIENCE

Software Engineer	May 2026 – Present
Verus Virtus	Kuala Lumpur, Malaysia
- Building Nexus, an AI-powered network device management platform that translates natural language into structured CLI commands for switch configuration, executed over SSH with full audit trail, plus a multi-vendor execution engine for non-native switch families via reviewed command templates - Setup an open source accounting system called BigCapital for the company's internal use using Nextjs, Postgres, and Tailwind CSS - Designed and built the company's landing page in Nuxt 3, Vue 3, and TypeScript, with scroll-pinned Three.js scenes (GSAP ScrollTrigger, Lenis) engineered for performance and accessibility - Involved in intelligence dashboard development comprises of 5 microservices using Nextjs, Postgres, and Shadcn+Tailwind CSS (NDA)	

Independent Software Engineer

Feb 2026 – Present

Self-Directed & Client Work

Malaysia

- Built and launched Rerumah (rerumah.my), a bilingual property-management PWA for Malaysian landlords on Next.js and Supabase, handling tenancy, billing, payment, and agreement workflows end-to-end
- Delivered a Malaysia Labour Market analysis dashboard for SOCSO using Power BI, building a multi-theme DAX semantic model across salary-gap and skills-demand analysis
- Collaborated with and delivered a landing page for an education agency on marketing coordination for a diploma programme's student acquisition campaign
- Currently in-progress collaborating and working on an MVP for an autonomous AI-agent agency for a client in the education sector utilizing Langchain libraries

Software Engineer (Security) – Short Contract

Feb 2026 – Apr 2026

Geomotion (Malaysia) Sdn. Bhd.

Malaysia

- Executed security hardening across 6 microservice repositories to remediate critical BHP audit findings, implementing API rate limiting, WAF ruleset configuration, and hardened HTTP security headers
- Strengthened CI/CD pipeline security by introducing automated code-scanning tools including Gype, GitLab SAST, and pip-audit across all deployment stages
- Implemented Content Security Policy (CSP) and Subresource Integrity (SRI) standards to strengthen web application security posture and prevent cross-site vulnerabilities

Software Engineer (Full-Stack)

Aug 2024 – Feb 2026

Silentmode Sdn. Bhd.

Malaysia

- Built the E-Invoice System (EIS) microservice for RONPOS from scratch using event-driven architecture on AWS (SQS, ECS, Lambda, DynamoDB, Kinesis), applying CQRS and Domain-Driven Design to process e-invoice submissions for LHDN integration; Vue.js frontend, NestJS backend
- Developed subsidy RON95 transaction and payment-processing features for the POS system with CDB team integration, handling 800K+ daily subsidy transactions across fuel stations nationwide using NestJS with DDD and TDD (Jest, MongoDB, Docker)
- Built the e-invoice submission module with creation, processing, and LHDN API submission, using message-queue patterns and event-driven architecture on AWS ECS
- Developed virtual loyalty-point integration in the receipt module and MyDebit card transaction support, collaborating with the Invenco team for the BHPetrol client
- Owned support tickets, hotfixes, and deployments for the HUB and EIS systems, ensuring high uptime across production environments

Junior Software Engineer / CloudBOS Developer

2023 – 2024

Terato Tech Sdn. Bhd. (outsourced to Silentmode)

Malaysia

- Developed multiple reporting features on CloudBOS, including fuel totalizer and sales-movement reports, deployed nationwide across Petronas, Shell, and BHPetrol stations, serving 1,000+ stations
- Designed and deployed an Electronic Shelf Label (ESL) system for in-store item purchases across nationwide fuel stations, with dynamic PDF/Excel generation
- Enhanced planogram generation to support dynamic customization (width, height, and custom input parameters) for fuel station owners
- Developed the frontend landing page for the KitaJaga website supporting seasonal campaigns, including the SemakHarga campaign

- Built the Betaqwa mobile app MVP from scratch using React Native and Expo, with end-to-end features including Search Masjid, Masjid Listing, Kuliah Listing, Masjid Details Profile, and Follow functionality
- Implemented a kuliah data-import feature in the Betaqwa CMS using Google Sheets API integration to optimize admin workflows
- Resolved critical bugs across frontend and backend modules, including email-delivery failures, modal rendering issues, and unhandled exceptions blocking user flows

PROJECTS

Nexus — AI Network Device Management Platform / *AI/LLM (Claude), SSH Automation, Multi-Vendor CLI Engine, Multi-Tenant SaaS, RBAC, WebSocket*

- AI-powered platform that translates natural language into structured CLI commands for network switch configuration (Aruba AOS-CX), executing over SSH with a full audit trail and confirmation-gated safety review
- Built a multi-vendor execution engine extending live operations beyond native AOS-CX to additional switch families via reviewed, template-driven command packs — the model selects a vendor-neutral operation while CLI output is sourced from stored templates, never generated freeform, with a permanent safety floor blocking catastrophic commands
- Implemented network discovery and snapshot grounding, doc-grounded AI reasoning via the Anthropic Files API, a browser-based SSH terminal (WebSocket/xterm.js), and a live DB-backed metrics dashboard
- Architected multi-tenant SaaS structure (Organizations → Brands → Locations → Devices) with role-based access control and a hybrid deployment model spanning cloud dashboard, offline/air-gapped local mode, and on-site agent

Verus Virtus — Company Landing Page / *Nuxt 3, Vue 3, TypeScript, Three.js, GSAP ScrollTrigger, Lenis, Tailwind CSS*

- Palantir-style marketing site for a deep-tech engineering firm, deployed at verusvirtus.com
- Scroll-pinned Three.js scenes on a shared single-canvas, single-renderer architecture, with lazy-loaded scene modules, mobile DPR capping, and reduced-motion fallbacks

E-Invoice System (EIS) / *NestJS, Vue.js, AWS ECS/SQS/Lambda/DynamoDB/Kinesis, LHDN API, Docker, CQRS, DDD, Event-Driven Architecture*

- E-invoicing solution for fuel retailers and end users, offered as part of RONPOS and integrated with the LHDN API
- Serverless microservice on AWS ECS implementing CQRS, DDD, and message-queue / event-driven patterns

HUB — Fuel POS Backend / *NestJS, MongoDB, Docker, MQTT, TypeScript, Jest, TDD, REST APIs*

- Backend for fuel-POS solutions covering transactions, receipts, pump integrations, payment integrations, and daily station operations
- Dockerized and deployed on edge-computing devices with isolated local MongoDB instances synchronized to cloud via MQTT, enabling offline-first operations across 200+ BHPetrol stations

CloudBOS — Back Office System / *Vue.js, Laravel, AWS Aurora (MySQL), AWS EC2, Multi-tenant Architecture, REST APIs*

- Multi-tenant reporting dashboard and configuration panels for the FuelPOS solution serving Petronas, with isolated data per client
- Fuel totalizer and sales-movement reports deployed nationwide across 1,000+ Petronas fuel stations

Rerumah — Property Management PWA / *Next.js, Supabase, PostgreSQL, Tailwind CSS, TypeScript, PWA, REST APIs*

- Bilingual (Malay-first) PWA for Malaysian landlords enabling end-to-end tenancy, billing, payment, and agreement management
- Lightweight, offline-first build (<2MB bundle) deployed at rerumah.my, supporting 50+ active properties in beta

Betaqwa — Mosque Management System / *Laravel, React Native, Expo, MySQL, AlpineJS, TanStack Query, Zustand, Google Sheets API*

- Laravel CMS plus a React Native (Expo) app with Masjid search, listings, kuliah schedules, and follow functionality
- Delivered a full MVP with open-source data for 500+ mosques across Malaysia

Malaysia Labour Market Dashboard (SOC SO) / *Power BI, DAX, Data Modeling, SQL*

- Processed 300K+ employment records with real-time refresh, analyzing 6 themes including salary-gap analysis and skills-demand mapping

EDUCATION

Universiti Kebangsaan Malaysia (UKM)
B.Sc. Software Engineering (Information Systems) — GPA 3.76, First Class Honours

2019 – 2023

CERTIFICATIONS

Certified Kubernetes Administrator (CKA)
Linux Foundation · Credential ID: LF-u1wvz7sxq4

Dec 2024

Certified DevOps Foundation
CCSD Council · Credential ID: 3e59299

Nov 2024